

Data Transformation Using Dplyr

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Data Manipulation

dplyr package

- We use the package `dplyr` to to data manipulation

Data Manipulation Verbs

- The verbs in dplyr package are given as:
 - i. Pick observations by their values `filter()`
 - ii. Reorder the rows `arrange()`
 - iii. Pick variables by their names `select()`
 - iv. Create new variables with functions of existing variables `mutate()`
 - v. Collapse many values down to a single summary `summarise()`
- The above verbs can be used alongside `group_by()`

How the verbs work

- All verbs work similarly:
 - i. The first argument is a data frame
 - ii. The subsequent arguments describe what to do with the data frame
 - iii. The results is a new data frame

Filter rows with `filter()`

- `filter()` allows you to subset observations based on their values.
- The first argument is the name of the data frame.
- The second and subsequent arguments are the expressions that filter the data frame.

Example

```
1 library(tidyverse)
2
3 # filter flights on january 1st
4
5 filter(flights, month == 1, day == 1)
6
7 # flights that departed in Nov or Dec
8
9 filter(flights, month == 11 | month == 12)
```


Arrange rows with `arrange()`

- `arrange()` works similarly to `filter()` except that instead of selecting rows, it changes their order.
- It takes a data frame and a set of column names (or more complicated expressions) to order by.

```
1 # arrange according to year, month and day
2 arrange(flights, year, month, day)
3
4 # arrange in descending order
5
6 arrange(flights, desc(dep_delay))
```

Select Columns with `select()`

- `select()` allows you to rapidly zoom in on a useful subset using operations based on the names of the variables.

```
1 select(flights, year, month, day)
```

- There are a number of helper functions here:
 - i. `starts_with("abc")` matches names that begin with “abc”.
 - ii. `ends_with("xyz")` matches names that end with “xyz”.
 - iii. `contains("ijk")` matches names that contain “ijk”.

Add new variables with `mutate()`

- `mutate()` always adds new columns at the end of your dataset so we'll start by creating a narrower dataset so we can see the new variables.

```
1 flights_sml <- select(flights,  
2   year:day,  
3   ends_with("delay"),  
4   distance,  
5   air_time  
6 )  
7 mutate(flights_sml,  
8   gain = dep_delay - arr_delay,  
9   speed = distance / air_time * 60  
10 )
```

Grouped Summaries with `summarise()`

- We can collapse a data frame into a single row using the `group_by` and `summarise` functions

```
1 # create data frame
2
3 Course <- c("BAS", "BAS", "BAS", "BST", "BST", "BST")
4
5 Marks <- c(23, 24, 21, 21, 28, 29)
6
7 student_data <- data.frame(Course, Marks)
8
9 # create the total score for each group
10
11 student_data <- student_data %>%
12   group_by(Course) %>%
13   summarise(Total = sum(Marks))
```

- We can get any statistical function inside the `summarise()` function.